

Vandna Gupta

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Professional Profile

M.Sc. Bioinformatics graduate (Chaudhary Charan Singh University, Completed with 86.25%) with an active research internship at ICMR-National Institute of Cancer Prevention & Research (NICPR), Noida. Specializes in end-to-end metatranscriptomics pipeline development, functional biomarker profiling using the KEGG database, NGS-based microbiome analysis, and multi-omics data interpretation in oral cancer research. Distinguished by advanced proficiency in frontier AI research tools — Claude AI (Anthropic), NeodiSC, AntiGravity AI, OpenCode AI, and Gemini CLI — enabling accelerated hypothesis generation, literature synthesis, and bioinformatics workflow design at a level uncommon among early-career researchers. Also experienced in molecular docking and protein-ligand simulation using AutoDock and AutoDock Vina. Certified in Advanced Microsoft Excel.

Key Achievements

- University Gold Medalist in M.Sc. Bioinformatics (86.25%), Chaudhary Charan Singh University
- Co-authored a systematic review and meta-analysis on oral microbiome biomarkers in OSCC — manuscript submitted to Elsevier
- Built a complete end-to-end metatranscriptomics pipeline (SRA Toolkit → FastQC → Trimmomatic → Bowtie2 → Kraken2/Bracken → STAR/HISAT2 → DESeq2/edgeR → KEGG) from scratch during internship
- One of very few early-career bioinformaticians integrating frontier AI tools (Claude AI, Gemini CLI, OpenCode) into active cancer genomics research workflows
- Developed an automated Python pipeline for extraction of BioSample IDs from research literature PDFs to streamline data collection for meta-analysis studies
- Certified in Advanced Microsoft Excel (2025) for data analysis and reporting

Education

Completed (86.25%)

M.Sc. Bioinformatics (Gold Medalist)

CCSU, Meerut

Completed

B.Sc. Biotechnology

CCSU, Meerut

Relevant Coursework: Genomics, Proteomics, Bioinformatics Algorithms, Molecular Biology, Computational Biology, Biostatistics

Research & Internship Experience

2025 – Present

Bioinformatics Research Trainee

ICMR-NICPR, Noida

Supervisor: Dr. Pramod Kumar (Scientist-D), Division of Molecular Biology

Main responsibilities and contributions:

- Designed and executed a complete metatranscriptomics pipeline for microbial community profiling from NCBI BioProject sequencing data, covering SRA retrieval, QC, host removal, taxonomic classification, RNA-seq alignment, differential expression, and KEGG functional interpretation
- Conducted functional profiling of cancer-associated biomarkers using the KEGG database to map metabolic and signaling pathways implicated in OSCC progression

- Analyzed NGS-derived microbiome datasets from oral and other cancer cohorts using Linux-based bioinformatics workflows in R and Python
- Applied statistical methods including Mann-Whitney and Kruskal-Wallis tests for differential abundance testing and multi-group comparisons
- Leveraged Claude AI (Anthropic) for AI-assisted literature review, rapid hypothesis generation, research documentation, and manuscript preparation
- Utilized NeodiSC and AntiGravity AI tools for specialized bioinformatics analysis and biomarker discovery workflows
- Generated publication-quality data visualizations — violin plots, boxplots, heatmaps — using ggplot2 and ComplexHeatmap in R
- Developed Python automation tools: PDF classifier (smoking vs. non-smoking papers), BioSample ID extraction pipeline from literature PDFs, AI-powered PubMed screening app for systematic review triage

2023 (May – Aug)

Summer Research Trainee

**Bebymil
International Pvt.
Ltd.**

- Carried out summer research training on "Basic Techniques of Molecular and Microbiology"
- Gained practical laboratory experience in core molecular biology protocols and microbiological techniques under professional supervision

Research Projects

Oral Microbiome Meta-Analysis in OSCC — Systematic Review (Manuscript Submitted)

- Co-authored systematic review analyzing oral microbiome composition across global OSCC patient cohorts; identified conserved biomarker taxa and assessed confounding impact of age and geographic variation
- AI-accelerated synthesis of 50+ peer-reviewed studies using Claude AI and OpenCode AI

Literature Mining Pipeline for Oral Host Microbiome

- Developed an automated pipeline for targeted paper searching on meta-transcriptomics in the oral host microbiome.
- Expanded the literature extraction framework to systematically identify and categorize papers related to OPMD and OSCC within metagenomic studies.

Metagenomics Pipeline — 16S rRNA Amplicon Analysis (QIIME2 + DADA2)

- Processed raw 16S rRNA amplicon sequencing data through a complete QIIME2 pipeline: paired-end import → DADA2 denoising → ASV table generation → taxonomic classification using SILVA database
- Generated diversity analyses (alpha and beta diversity), taxonomic bar plots (phylum to species level), and differential abundance outputs to characterize oral microbial community shifts in OSCC vs. healthy controls
- Produced publication-quality taxonomic bar plots and PCoA plots to visualize community composition across patient cohorts

Functional Biomarker Profiling via KEGG Database

- Mapped cancer-associated microbial taxa to KEGG metabolic and signaling pathways using DADA2 ASV pipeline, 16S rRNA profiling, and QIIME2
- Integrated Claude AI-assisted annotation with statistical outputs for accelerated pathway-level biological interpretation

Molecular Docking & Simulation — AutoDock / AutoDock Vina

- Performed protein-ligand molecular docking using AutoDock and AutoDock Vina to evaluate binding affinities and interaction modes of candidate bioactive compounds against cancer-relevant target proteins

- Prepared receptor and ligand structures (PDB processing, energy minimization), defined grid boxes, and analyzed docking poses using PyMOL and Discovery Studio Visualizer
- Applied GROMACS and NAMD for molecular dynamics simulations to assess stability of docked complexes over simulation trajectories

Metabolomics ML Classification Pipeline — Control vs. OSCC

- Developed ML classification pipeline using XGBoost, Random Forest, SVM, and Neural Networks (scikit-learn, TensorFlow) for metabolomics data analysis

Microbiome Statistical Analysis & Visualization

- Performed differential abundance testing of microbial taxa in OSCC vs. healthy controls; reproduced diverging orange-blue heatmap of cellular biomarkers in R (ComplexHeatmap) and Python (seaborn/matplotlib)

Technical Skills

OS & Shell	Linux (Ubuntu) — command line, shell scripting, pipeline execution
Languages	Python, R, Bash
Metatranscriptomics	SRA Toolkit, FastQC, Trimmomatic, Bowtie2, Kraken2, Bracken, STAR, HISAT2, DESeq2, edgeR
Metagenomics	QIIME2, DADA2, 16S rRNA amplicon analysis, taxonomic bar plots, alpha/beta diversity, SILVA database
Molecular Docking	AutoDock, AutoDock Vina, PyMOL, Discovery Studio Visualizer — protein-ligand docking & binding affinity analysis
MD Simulations	GROMACS, NAMD — molecular dynamics simulation, trajectory analysis, complex stability assessment
Machine Learning	XGBoost, Gradient Boosting, Random Forest, SVM, Neural Networks, Scikit-learn, TensorFlow
AI Research Tools	Claude AI (Anthropic), NeodiSC, AntiGravity AI, OpenCode AI, Gemini CLI
Databases	KEGG, NCBI, BioProject, ClinVar, SILVA, PDB, genomic databases
Visualisation	ggplot2, ComplexHeatmap, seaborn, matplotlib, QIIME2 plots, PyMOL
Data & Reporting	Advanced Microsoft Excel (Certified), Jupyter Notebook, RStudio
Statistics	Mann-Whitney, Kruskal-Wallis, differential expression analysis, meta-analysis

Certifications & Training

2025	Advanced Microsoft Excel	E-Certificate
2025–Present	Bioinformatics Research Internship	ICMR-NICPR, Noida
2023	Basic Techniques of Molecular and Microbiology	Bebymil International Pvt. Ltd.